Unlock the code: acceleration

07 C

DIRECTIONS · Solve each station. Copy the letter beside your answer into the lock.

STATION 1

A cart goes from 0 to 8 m/s in 4 s. Find a.

(M) 4 m/s² (C) 2 m/s² (R) 32 m/s²

STATION 2

Velocity drops from 12 to 4 m/s in 2 s. Find a.

(P) +4 m/s² (T) −8 m/s² (H) −4 m/s²

STATION 3

v = +5 m/s and a = −2 m/s². The object is...

(E) speeding up (A) slowing down

STATION 4

v = −6 m/s and a = −3 m/s². The object is...

(L) slowing down (N) speeding up

STATION 5

v_0 = 2 m/s, a = 3 m/s², t = 4 s. Use $v = v_0 + at$.

(D) 12 m/s (G) 14 m/s (S) 9 m/s

STATION 6

A car goes 0 → 27 m/s in 6 s. Find a. ($g \approx 9.8$ m/s²)

(O) 162 m/s² (E) 4.5 m/s² (U) 21 m/s²

LOCK [1] [2] [3] [4] [5] [6]

Hint: what acceleration measures in velocity.

R4 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 7 · R4 · SEI · PARTNER TALK

Talk it through: acceleration

Language goal: I can **explain** speeding up and slowing down.

DIRECTIONS · Partner A and B read their lines aloud. Listen, then write. Switch each round.

ROUND 1 · A CAR: 0 → 12 m/s IN 4 s



0 → 12 m/s in 4 s

A ASKS "What is the acceleration of the car?"

B SAYS "The change in velocity is ____. I divide by ____. So a = ____."

A CHECKS "So you're saying the car gains ____ m/s every second. I agree / disagree because ____."

ROUND 2 · SWITCH · A BIKE BRAKES



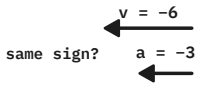
bike brakes

B ASKS "Is the bike speeding up or slowing down? How do you know?"

A SAYS "The bike is ____ because the gaps ____. Its acceleration points ____ the motion."

B ADDS "I want to add that braking is still acceleration because ____."

ROUND 3 · BOTH · v = −6 m/s, a = −3 m/s²



A "I think it is ____ because the signs are ____."

B "I agree / I see it differently. The minus sign means ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

Acceleration means…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

R4 (1) (2) (3)

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Settle the argument

07 I

DIRECTIONS · Shade the bubble of the person who is right. Give evidence, then fix the wrong claim.

ROUND 1

ANA
"Negative acceleration means slowing down."

BEN
"Not always — it depends on the sign of the velocity too."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 2

KAI
"A car going a steady 20 m/s on a curve isn't accelerating."

LU
"It is — its direction, so its velocity, is changing."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 3

MAYA
"2 → 10 m/s in 4 s gives a = 10 ÷ 4 = 2.5 m/s²."

RAJ
"You need the change: a = (10 − 2) ÷ 4."

Who's right? _____ · Evidence: _____

Fixed claim: _____

YOUR TURN · WRITE A TRAP

Write a claim about today's idea that sounds right but is wrong. Trade and settle it.

R4 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Say it, then write it

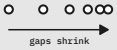
Language goal: I can **explain** acceleration with **because**.



accelerate
acelerar · acelerar
to change velocity



speed up
acelerar · acelerar
go faster · gaps grow



slow down
frenar · desacelerar
go slower · gaps shrink



change in
cambio de · variação de
Δ = end − start

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: faster · slower · velocity · second · change

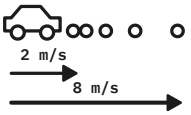
Acceleration tells how much the ____ changes each ____. When a car goes ____, it speeds up. When it goes ____, it slows down. Both are a ____ in velocity.

2 · SAY IT TO A PARTNER, THEN WRITE IT



The car is ____ because the gaps ____ .
v and a have ____ signs, so the car is ____ .

3 · SOLVE WITH THE STEPS



A bike goes from 2 m/s to 8 m/s in 3 seconds.

STEP 1 · CHANGE
____ − ____ = ____ m/s

STEP 2 · DIVIDE
____ ÷ ____ = ____ m/s²

STEP 3 · SAY IT
"The bike gains ____ m/s every second."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ Braking is acceleration because ____ .
☐ I know the car speeds up because ____ .
☐ First the velocity is ____, then ____, so a is ____ .

I said it ☐ · I wrote it ☐

R4 1 2 3

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

100 m final – the first 5 seconds

● LIVE

You're the broadcast analyst. Made-up practice race.

07 X

SPLIT DATA

t (s)	v (m/s)	a (m/s²)
0	0	–
1	4.0	
2	7.5	
3	10.0	
4	11.0	
5	11.0	

SLOW-MO v–t GRAPH

STAT 1 · BIGGEST a

Which second? ____

STAT 2 · a REACHES 0

When? ____ Still moving? ____

STAT 3 · AVERAGE a, 0–4 s

a = ____ · vs. g ≈ 9.8: ____

COMMENTARY BOOTH

"Out of the blocks she accelerates at ____ – a gain of ____ every second! By second 4 she holds a top speed of ____, so her acceleration is ____ … but she is still ____!"

R4 1 2 3

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 7 · R4 · LOOP CHAIN

I have... who has? Acceleration

07 G

DIRECTIONS · Start at ★. Solve, find the tile that "has" your answer, draw an arrow, repeat.

★ START — **A**

WHO HAS…

0 → 8 m/s in 4 s. What is a?

I have **14 m/s** — **B**

WHO HAS…

0 → 27 m/s in 6 s. What is a?

I have **−4 m/s²** — **C**

WHO HAS…

v = +5, a = −2. Speeding up or slowing?

I have **m/s²** — **D**

WHO HAS…

v = −6, a = −3. Speeding up or slowing?

I have **2 m/s²** — **E**

WHO HAS…

12 → 4 m/s in 2 s. What is a?

I have **speeding up** — **F**

WHO HAS…

Steady 20 m/s around a curve. Accelerating?

I have **slowing down** — **G**

WHO HAS…

v0 = 2, a = 3, t = 4 s. What is v?

I have **4.5 m/s²** — **H**

WHO HAS…

The units of acceleration?

I have **yes — direction changes** — **I**

END

MY PATH: A → ___ → ___ → ___ → ___ → ___ → ___ → ___ → END

R4 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else